

Advanced (Carolina Reaper)

Q1. What is the unit rate of 2 apples for

1.50

?

- (A) 1.00 per apple
- (B) 1.25 per apple
- (C) 1.50 per apple
- (D) 2.00 per apple
- (E) 3.00 per apple

Q2. A car travels 240 miles in 4 hours. What is its speed in miles per hour?

- (A) 40 mph
- (B) 50 mph
- (C) 60 mph
- (D) 70 mph
- (E) 80 mph

Q3. If y varies directly with x , and $y = 12$ when $x = 4$, what is the equation of variation?

- (A) $y = 2x$
- (B) $y = 3x$
- (C) $y = 4x$
- (D) $y = 5x$
- (E) $y = 6x$

Q4. A bakery sells 480 loaves of bread in 8 days. What is the rate of loaves sold per day?

- (A) 40 loaves/day
- (B) 50 loaves/day
- (C) 60 loaves/day
- (D) 70 loaves/day
- (E) 80 loaves/day

Q5. If z varies inversely with w , and $z = 15$ when $w = 3$, what is z when $w = 9$?

- (A) 5
- (B) 10
- (C) 15
- (D) 20
- (E) 25

Q6. What is the unit rate of 3 books for

12

?

- (A) 2 per book
- (B) 3 per book
- (C) 4 per book
- (D) 5 per book
- (E) 6 per book

Q7. A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, how many pieces of candy will each friend get?

- (A) 4 pieces
- (B) 5 pieces
- (C) 6 pieces
- (D) 7 pieces
- (E) 8 pieces

Q8. A water tank can hold 1200 gallons of water. If 300 gallons of water are already in the tank, what percentage of the tank is filled?

- (A) 20
- (A) 25
- (A) 30
- (A) 35
- (A) 40

Open Ended Questions (FRQ)

Q9. A car rental company charges a base fee of

20

plus an additional

0.25

per mile. If a customer rents a car for a day and drives x miles, what is the total cost $C(x)$ in dollars? Show your work and simplify your answer.

Q10. A bakery is having a sale on bread. A regular loaf of bread costs

2.50

, but the bakery is offering a 10

Chef's Tips

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- Make sure students show their work and simplify their answers for the free response questions.
- Emphasize the importance of using proper units and notation.
- Encourage students to use calculators to check their work, but only if they have already shown their work and arrived at an answer.